Replication Project

Can Warm Behaviour Mitigate the Negative Effect of Unfavourable Governmental Decisions on Citizens' Trust?

Replication Project

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Paper Theory

- Citizens' trust is fundamental in successful governance
- Trust is often based on performance (e.g. economic stability) and decisions that fit their policy preferences
- There is some evidence that direct interactions with bureaucrats (e.g. applying for benefits) impact broader government trust
- From social psychology, we understand the importance of impressions when making evaluative judgements
- When citizens interact with government employees, what is the impact on the perceived traits of this employee and does this impact broader government trust?

H1: Citizens have lower trust in government the more unfavourable a decision is to the individual.

H2: The effect of decision favourability on citizens' trust in government depends on whether bureaucrats show warmth in the decision-making process. The negative effect of unfavourable decisions is weaker when bureaucrats show high warmth compared to showing low warmth.

Method

- 2 x 2 survey experiment on representative sample of Danish Citizens (n = 1613)
- Participant read a vignette in which they are helping a close elderly relative apply for a place in a nursing home
- Vignette manipulates two features: the description of the caseworker (high or low warmth) and decision on the application (accepted versus rejected).
- Participants answered a series of demographic questions as well as rating the caseworker and decision on a number of traits
- To record the dependent measures they rated on a 0-10 scale how much they trusted (a) the bureaucrat and (b) the municipality

Government Caseworker

Result of Nursing Home Application	Warmth = 1	Warmth =0
Favourability = 0	Group 1 High Warmth + Unfavourable Decision	Group 4 Low Warmth + Unfavourable Decision
Favourability = 1	Group 2 High Warmth + Favourable Decision	Group 3 Low Warmth + Favourable Decision

i.e. "the caseworker answered all your questions in a [warm and friendly / cold and unfriendly] manner … "
"based on the caseworker's recommendation, the municipality has decided to [reject/approve] the application to enter a nursing home.

"How much do you personally trust the [municipality/the caseworker] to make the right decisions?"

0	1	2	3	4	5	6	7	8	9	10

'No Trust At All' (0) — 'Completely Trust' (10)

Social- og Boligstyrelsen
Viden til gavn

caseid	treatment_groups	of_treatment	warmth_treatment	trust_b	trust_m	gender
1818592968	3	1	0	0.5	0.0	0
1823485372	1	0	1	0.3	0.3	1
1823435322	1	0	1	0.3	0.3	1
1823435332	4	0	0	0.2	0.2	1
1823492020	3	1	0	NA	NA	0
1821707542	1	0	1	0.6	0.5	0
1823492796	4	0	0	0.7	0.7	1
1823492886	1	0	1	0.5	0.7	1
1823373596	3	1	0	0.8	0.8	0
1823493046	3	1	0	0.0	0.0	0

Data

- The two experimental treatments were coded into dummy variables: ’
    - of treatment $\rightarrow$ 0 = Unfavourable, 1 = Favourable
    - warmth treatment $\rightarrow$ 0 = Low Warmth, 1 = High Warmth
- Outcome variables ’trust b’ and ’trust m’ were rescaled from their original ‘0-10’ scale to a ‘0-1’ scale
    - E.g. 5 $\rightarrow$ 0.5
- Other demographic variables and manipulation validity test measures were included

Included Variables

```
 [1]  "caseid"             "user"
[3]  "pubsector_pref"     "treatment_groups"
[5]  "q5"                 "fair"
[7]  "neutral"            "unbiased"
[9]  "warm"               "friendly"
[11]  "likable"            "competent"
[13]  "intelligent"        "capable"
[15]  "gender"             "age"
[17]  "education_rc"       "ideology"
[19]  "age_1"              "edu"
[21]  "attention"          "trust_m"
[23]  "trust_b"            "of_treatment"
[25]  "warmth_treatment"   "perc_warmth"
[27]  "perc_fairness"      "perc_competence"
[29]  "perc_favorability"
```

Original Study

- The author conducts four OLS linear regression, with the two trust outcomes variables testing simple and interaction models
- Decision favourability significantly increases citizens level of trust in both the bureaucrat and the municipality, an increase of 18.8% points and 17.9% points respectively → supporting H1
- The effect of outcome favourability is not weaker when bureaucrats showed high warmth versus low warmth, as demonstrated by the non-significant interaction terms for both outcome → not supporting H2

Table 2: OLS Experimental Results

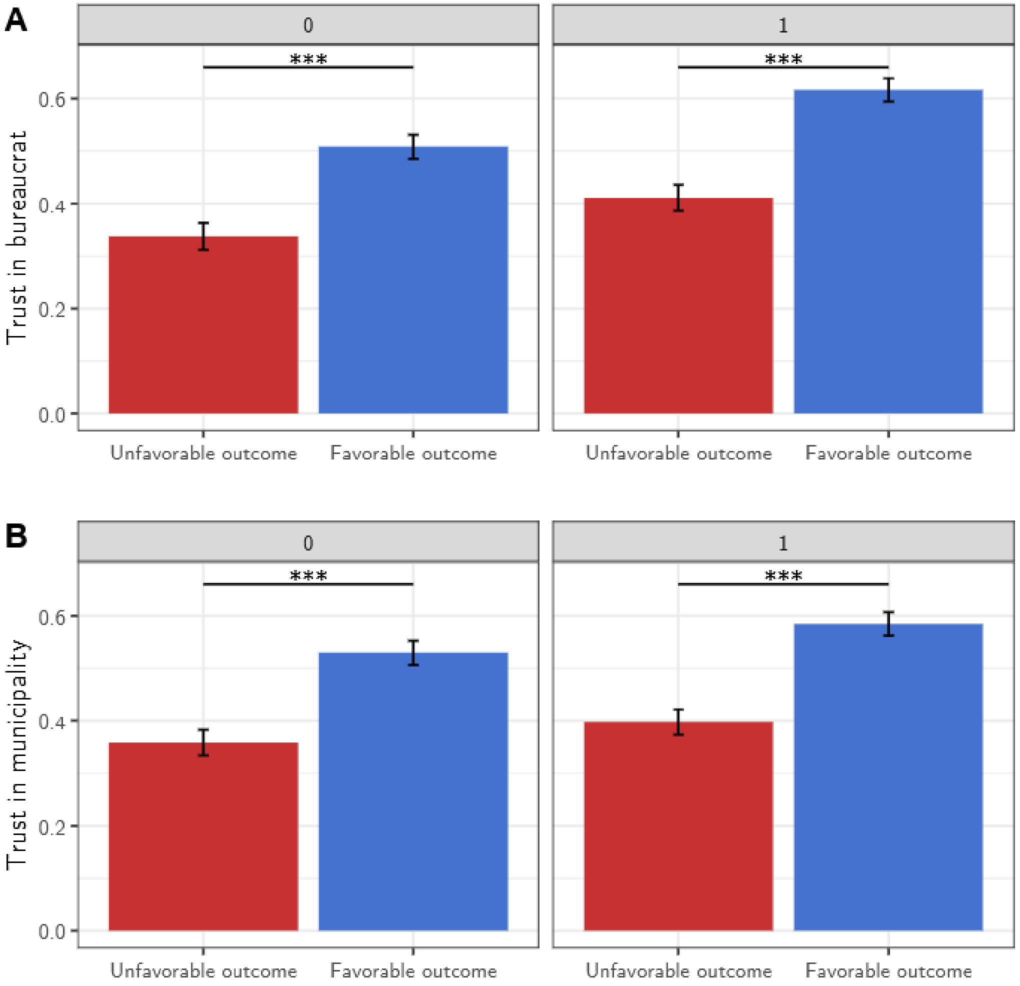
	Dependent variable:			
	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat	Trust Municipality
	(1)	(2)	(3)	(4)
Outcome favorability (OF)	0.188***	0.179***	0.171***	0.171***
	(0.012)	(0.012)	(0.017)	(0.017)
Warmth	0.091***	0.047***	0.073***	0.039**
	(0.012)	(0.012)	(0.017)	(0.017)
OF * Warmth			0.035	0.016
			(0.024)	(0.024)
Constant	0.329***	0.355***	0.338***	0.359***
	(0.011)	(0.010)	(0.012)	(0.012)
Observations	1,516	1,525	1,516	1,525
R^2	0.163	0.136	0.164	0.137
Adjusted R^2	0.162	0.135	0.163	0.135
Residual Std. Error	0.237 (df = 1513)	0.233 (df = 1522)	0.236 (df = 1512)	0.233 (df = 1521)
F Statistic	147.665*** (df = 2; 1513)	120.190*** (df = 2; 1522)	99.197*** (df = 3; 1512)	80.248*** (df = 3; 1521)
Note:				*p<0.1; **p<0.05; ***p<0.01

Original Study

- Warmth significantly increases the level of trust in the bureaucrat and the municipality, both for both unfavourable and favourable conditions
- The Favourable + High Warmth group had the largest average trust scores of 0.617 and 0.585 respectively
- Hansen concludes that "*public leaders should be concerned about whether their employees behave in a warm manner if they want to build greater trust*"

Experimental Group	Trust b	Trust m
Unfavourable / Low Warmth	0.338	0.359
Unfavourable / High Warmth	0.411	0.398
Favourable / Low Warmth	0.508	0.530
Favourable / High Warmth	0.617	0.585

Figure 1: Interaction Estimates



Notes: Effect of the outcome favourability manipulation on trust in the bureaucrat (Panel A) and trust in the municipality (Panel B) conditional on the warmth treatment. Bars are means and lines are 95% CIs. $***p < 0.001$..

Ordered Multinomial Regression

- Potential issue of OLS assuming the outcome variable is continuous and unbounded when the 0-10 scale is technically both ordinal and bounded
- Four ordered multinomial regression models were run, treating the original outcomes levels 0-10 as separate categories on an ordinal scale and the coefficients refers to whether the log-odds of being in a higher category changes in each treatment.
- The positive impact of the treatments and insignificant interaction term are consistent with the OLS models, and the coefficients are interpreted in terms of log-odds.
  - E.g. For model 1 and holding all else constant, participants in the warmth condition have on average about 2 times higher odds of choosing a higher trust category than those not in the warmth treatment ($OR = e^{0.692} \approx 1.99$)
- Although the interpretation is slightly less intuitive than OLS, this approach does make movement across the ordinal responses clear
- To obtain more interpretable results, the predicted probabilities of categories for each treatment group was calculated and then converted into an expected trust measure.
- The results clearly show the differences between predicted average trust scores for each group that are reasonably similar to the actual experimental group mean suggesting that the ordered regression is reasonably well fit
  - E.g. For Unfavourable Low Warmth the Trust B in the simple model predicted mean trust score was 3.25 compared to the observed 3.38.

Table 5: Ordered Multinomial Regression Coefficients

	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat	Trust Municipality
of_treatment1	1.409 (0.096)***	1.350 (0.095)***	1.265 (0.131)***	1.304 (0.131)***
warmth_treatment1	0.692 (0.091)***	0.333 (0.090)***	0.545 (0.129)***	0.288 (0.127)*
of:warmth			0.291 (0.181)	0.091 (0.179)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Figure 2: Ordered Multinomial Regressions: Predicted Trust

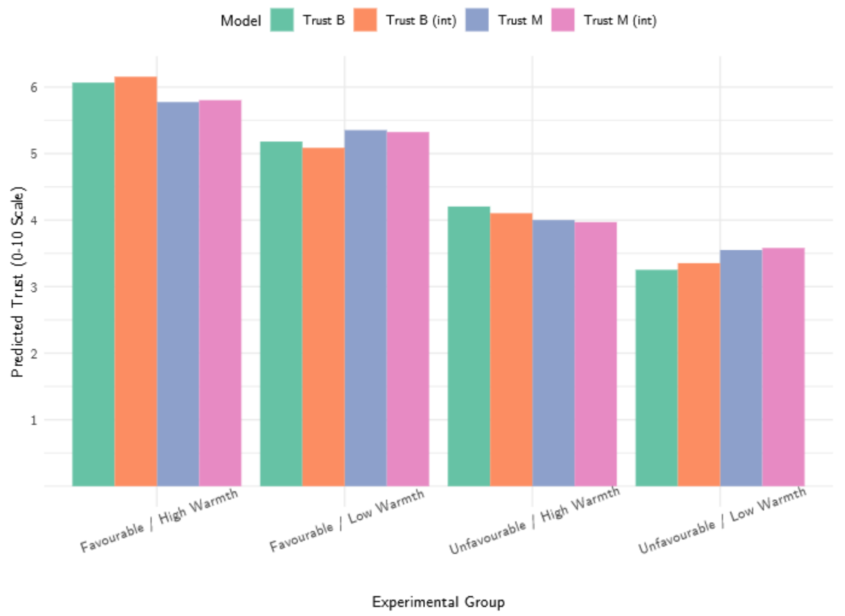


Table 3: Predicted Trust by Experimental Group and Model

	Group	Trust B	Trust M	Trust B (int)	Trust M (int)
1	Unfavourable / Low Warmth	3.25	3.55	3.35	3.58
2	Favourable / Low Warmth	5.18	5.35	5.09	5.32
3	Unfavourable / High Warmth	4.20	4.00	4.10	3.97
4	Favourable / High Warmth	6.07	5.78	6.16	5.80

Ordered Beta Regression

```r
library(cmdstanr)
library(brms)
library(ordbetareg)
cmdstanr::cmdstan_path()
set.seed(123)

reg_beta_b <- ordbetareg::ordbetareg(trust_b ~ of_treatment + warmth_treatment
    , data = data)
reg_beta_m <- ordbetareg::ordbetareg(trust_m ~ of_treatment + warmth_treatment
    , data = data)
int_beta_b <- ordbetareg::ordbetareg(trust_b ~ of_treatment * warmth_treatment
    , data = data)
```

- Ordered Beta regression is an alternative type of regression explicitly designed for outcome data that has an ordinal and bounded structure.
- It combines beta regression defined for any bounded continuous scale and the ordered logit that uses discrete categories.
- It is beneficial as it produces intelligible estimates that respect the bounds of the outcome variable as it assumes there is an unobserved latent scale that is mapped with ordinal cut-offs for the scaled ratings.
- The package 'ordbetareg' estimates a Bayesian implementation (Markov chain Monte Carlo (MCMC) finding the posterior distribution
- The predicted mean trust values for the groups were calculated from posterior predictions and show very similar prediction to the real means

Figure 3: Ordered Beta Regressions: Predicted Mean Trust Values by Group

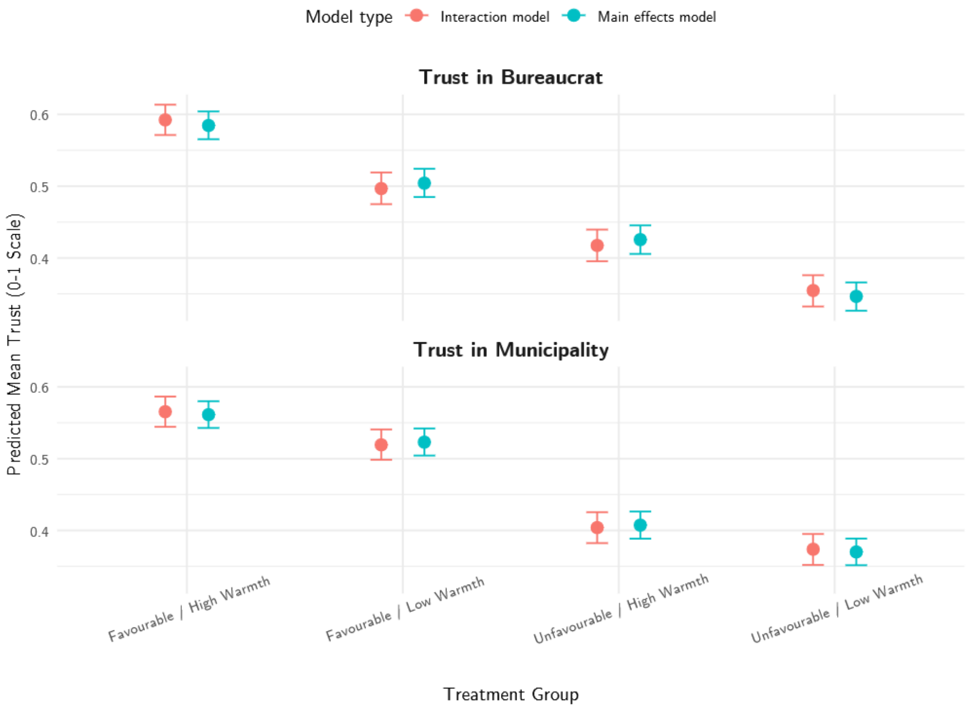


Table 8: Ordered Beta Regression: Predicted Trust of Experimental Groups with Confidence intervals

treatment	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat (int)	Trust Municipality (int)
Unfavourable / Low Warmth	0.35 [0.33, 0.37]	0.37 [0.35, 0.39]	0.35 [0.33, 0.38]	0.37 [0.35, 0.40]
Unfavourable / High Warmth	0.43 [0.41, 0.45]	0.41 [0.39, 0.43]	0.42 [0.40, 0.44]	0.40 [0.38, 0.43]
Favourable / Low Warmth	0.50 [0.48, 0.52]	0.52 [0.50, 0.54]	0.50 [0.47, 0.52]	0.52 [0.50, 0.54]
Favourable / High Warmth	0.58 [0.57, 0.60]	0.56 [0.54, 0.58]	0.59 [0.57, 0.61]	0.57 [0.54, 0.59]

Marginal Effects

Table 9: Marginal Effects of Warmth Treatment by OF Treatment and Model

of_treatment	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat (Interaction)	Trust Municipality (Interaction)
0	0.16 [0.14, 0.18]	0.15 [0.13, 0.17]	0.14 [0.11, 0.17]	0.15 [0.12, 0.17]
1	0.16 [0.14, 0.18]	0.15 [0.13, 0.17]	0.17 [0.15, 0.21]	0.16 [0.13, 0.19]

- Marginal Effects of Warmth calculated to understand how much increasing warmth (from Low Warmth to High Warmth) changes predicted trust and whether this depends on the outcome favourability.
- The values in the table represent the marginal effect of warmth on the predicted trust (0-1) scale returned by the ordered beta regressions.
- These show that participants receiving the unfavourable decision (when of treatment = 0) the effect of the warmth treatment increases trust in the bureaucrat and the municipality by 0.16 and 0.15 points on average, on the 0-1 scale (for non-interaction models). This marginal effect is exactly the same when participants receive the favourable decision (of treatment = 1).
- This suggests that warmth has almost exactly the same effect on trust in the favourable and unfavourable condition providing no evidence that there is a different effect depending on the type of decision.
- This coincides with the insignificant finding of the interaction term for the previous models

Conclusion

- The original study was replicated successfully.
- Firstly, I assessed an ordinal multinomial which was designed to reflect the ordinal nature of the data.
- This showed similar results to the OLS models, made similar predictions and further supported the lack of significant interaction.
- Secondly, I ran an ordered beta regression as it respected both the ordinal and bounded structure of the rescaled outcome measures.
- This approach reaffirmed the results from the original OLS study, finding a consistent marginal effect of warmth for both outcome conditions, but with an approach that is specifically designed for bounded ordinal nature of the outcome variable.

Thank you for listening!

Any Questions?